Universitat de Barcelona

Ricci Collineations of Plane-Symmetric Space Times

Friederike Krüger

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Supervisor of the Masterthesis: Josep Llosa

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1 Introduction

The importance of symmetries in physics has an outstanding history. The principles of modern theoretical physics in most of their aspects are based on the exploration of the symmetries we find in nature. They give information on conservation laws, following *Noether's theorem*, which states that to every symmetry there is a conserved quantity. Hence, [1] in the theory of general relativity space time symmetries are of great importance. They play an important role in the understanding of the relation between space time and matter, since they help examining exact solutions of the highly non-linear Einstein Equations.

A so called **symmetry of space time** [2] is the property of invariance: it is a certain set of transformations that preserves the structure of the space time. Thus, it is a mapping of the object onto itself. A space time symmetry is mathematically manifested in the vanishing of the Lie derivative of the corresponding object along the flow of a vector field that preserves the symmetry. It is an infinitesimal transformation which leaves the object invariant. The vector field carrying this property is called the infinitesimal generator.

An **isometry** [3] is a special kind of symmetry: it is a distance preserving mapping from a metric space to another that is bijective. The generators of isometries of space time are the **Killing vector fields**, characterised as the vector fields along which the Lie derivative of the metric vanishes. For example, the isometries in Minkowski space time are the transformations of the Poincaré group: rotation, translation, reflection and boost. They build a ten-dimensional Lie group and are well studied in special relativity. In general one can find whether there are symmetries as spatial and time translation or rotations by solving the Killing equation.

A **conformal transformation** is a mapping that preserves the angles. This [4] is very useful for the understanding of the relations between theories of gravity and Einstein's relativity. Conformal transformations of the metric play a role in scalar-tensor theories of gravity. The use of conformal transformations offers a different frame to discuss physical processes in general relativity as inflation or density perturbations and can simplify the analyzation of certain problems, for example by compactifying a formally infinite metric into a compact interval. This gives raise to understanding the space time through Penrose diagrams.

In general relativity **geodesics** are the trajectories of free point particles or they motion just due to gravity, i.e. their motion through curved space. So describe timelike geodesics the motion of a free falling particle. **Affine transformations** are the space time symmetries that preserve the geodesics.

The **Riemann tensor** is an invariant measure of curvature. It's a quantity derived from the commutation of the covariant derivative that attaches a tensor to each point in space that is a measure of curvature. It includes second derivatives of the metric, which give information about the curvature. The first covariant derivative of the metric corresponds to the Christoffel symbols, but since one can in any space time always find a coordinate system in which the Christoffel symbols are 0, first derivatives of the metric are not sufficient to gain information about the curvature. Moreover, Christoffel symbols are no tensors and hence not invariant under coordinate transformations. The Riemann tensor on the other hand does not depend on the choice of coordinates. One can picture it by taking an infinitesimal closed path about an infinitesimal region on the manifold and parallel transport a vector halfway along this path to a point and halfway to the other direction of the path to the same point. The Riemann tensor measures the difference of the direction of the vector depending on the way it was parallel transported along. It gives the curvature of the region of this infinitesimal path. If the Riemann tensor is 0 this means that there is no curvature in this region.

There are other, physically significant symmetries called **collineations**, such as Riemann collineations, Ricci collineations or Matter collineations, that are characterized through the vanishing of the Lie derivative of the Riemann tensor, the Ricci tensor or the Stress-Energy tensor, respectively, along the collineation vector field.

The **Ricci tensor** is the first contraction of the Riemann tensor, the Ricci scalar is the total contraction of the Riemann tensor. They are important objects to understand the geometry of the metric. Both of them are related to the Stress-Energy tensor over the Einstein Field Equations, that describe the fundamental interaction of gravity. Matter collineation are the symmetries of the Stress-Energy tensor. They are of special interest since they are related to conservation laws on matter fields [1]. Every Killing vector field is a Riemann, Ricci and affine collineation field, since these tensors are build out of the metric, but, the other way around, the latter ones do not need to preserve the metric. Hence collineation fields are a generalization of Killing vector fields.

Ricci collineation fields are vector fields that preserve certain symmetries of the Ricci tensor, similarly as Killing vector fields preserve symmetries of the metric. In general, collineation fields X fulfil the condition

$$\pounds_X T = 0 \tag{1}$$

where T denotes a symmetric covariant 2-tensor and $\mathcal{L}_X$ the Lie derivative along X.

Specifying for the case of the Ricci tensor we have

$$\pounds_X R_{\mu\nu} = 0. \tag{2}$$

If $R_{\mu\nu}$ is a diagonal non-degenerate tensor we can take it as a "new metric" and eq. (2) is like a usual Killing equation. However, if $R_{\mu\nu}$ is generate we need to use other techniques to find the collineation fields and their class might be non-finite dimensional. Also they might not obey a Lie algebra.

We shall here study the Ricci collineations for the specific class of static plane symmetric space times which has been considered elsewhere [5] merely by brute force, aiming to test the methods and classification of Ricci collineations developed in [6] and prove that they provide a powerful tool that permits a more elegant and systematic approach.

2 Theory of the Killing Equation

2.1 In n Dimensions

If the Ricci tensor, here called T_{ab}, is diagonal and has the full rank $\{rank(T) = n\}\,; detT \neq 0$ we can take this tensor as a non-degenerate metric tensor and equation (1) for the Ricci collineations can be seen as a usual Killing equation. Hence we can find the Ricci collineation field by solving the Killing equation.

$$\begin{aligned} \pounds_X T_{ab} \equiv \nabla_a X_b + \nabla_b X_a = 0 \\ \Leftrightarrow \partial_a X_b + \partial_b X_a - 2\Gamma^c_{ab} X_c = 0 \end{aligned} \tag{3}$$

where the covariant derivative and the Christoffel symbols are the Levi-Civita connection for what we call the new metric tensor T_{ab} and which is actually the Ricci tensor of the metric we start with. Eq. (3) implies that $\nabla_a X_b$ is a skewsymmetric tensor Ω_{ab}

$$\nabla_a X_b = \Omega_{ab} \qquad\qquad \Omega_{ab} + \Omega_{ba} = 0 \tag{4}$$

However the skew symmetric tensor Ω_{ab} cannot be arbitrarily chosen because equation (4) carries some integrability conditions that follow from the Ricci identity [7]

$$\nabla_b \nabla_c X_a - \nabla_c \nabla_b X_a = X_d R^d{}_{abc}. \tag{5}$$

Now, as $\Omega_{ab} = \nabla_{[a} X_{b]}$, it fulfils a Jacobi identity $\nabla_a \Omega_{bc} + \nabla_b \Omega_{ca} + \nabla_c \Omega_{ab} = 0$ and combining these two relations we finally arrive at

$$\nabla_a \Omega_{bc} = X_d R^d{}_{abc}. \tag{6}$$

From this equation (6) it follows that [8]

$$\pounds_X \Gamma^c_{ab} \equiv \nabla_a \nabla_b X^c - X^d R_{dab}{}^c = 0. \tag{7}$$

Hence equation (4) amounts to the fact that X^a preserves the metric and equation (6) is equivalent to the preservation of the Levi-Civita connection.

Equations (4) and (6)are a complete partial differential system whose general solution, if it exists, depends on the parameters $X_a(x_0)$ and $\Omega_{ab}(x_0)$ where x_0 is an initial point. The class of Killing vectors is thus a vector space with a finite number of dimensions $\begin{pmatrix} \leq n + n \\ 2 \end{pmatrix}$

However eq.(7) is subject to integrability conditions as well. It could be derived from the Ricci identities or, equivalently, requiring that all geometric objects derived from the metric tensor T_{ab} are preserved by X_a as well, namely

$$\begin{aligned} \pounds_X R_{abcd} = 0 \\ \pounds_X \nabla_{e_1} R_{abcd} = 0 \\ n = 0, 1, 2... \qquad \pounds_X \nabla_{e_1} ... \nabla_{e_n} R_{abcd} = 0 \end{aligned} \tag{8}$$

This leads to an infinite hierarchy of equations that constrain the parameters $X_a(x_0)$ and $\Omega_{ab}(x_0)$ in the partial differential system (4), (7). For example, the first tensor equation of (8) gives 20 independent component equations, the second one 56 equations and so on. These equations are a subsidiary system of constrains on the initial data. They are taken as point equations and all of them have to be fulfilled at x_0 in space time.

However, we will use another approach, exploring the fact, that all equations of (8) are contained in the first of the equations, the Lie derivative of the Riemann tensor, under the condition of taking it not as a point, but as a "function equation" that is fulfilled everywhere in space time. We will take this equation as the main algebraic system, we will modify it, find a general solution by algebraic methods and then require that it fulfils equations (3) as well. If we are in $n \geq 4$ dimensions it makes sense to modify our system by decomposing the Riemann tensor in its scalar, semi-traceless and fully-traceless part. Instead of working with the Riemann tensor directly we will proceed with the Ricci and Weyl tensor.

$$\pounds_X R_{abcd} = 0 \qquad \Leftrightarrow \qquad \pounds_X R^a{}_b = 0 \qquad \pounds_X C^{ab}{}_{cd} = 0 \tag{9}$$

with R^a_b being the Ricci tensor with a raised index and $C^{ab}{}_{cd}$ being the Weyl tensor with the first 2 indices raised. With the two latter equations we can construct all scalar invariants,i. e. the traces of all products of R^a_b with itself up to n times (resp. $C^{ab}{}_{cd} \quad m = \binom{n}{2}$). The scalar invariants of the Riemann tensor will just be algebraic functions of the scalar invariants we constructed. We will denote the scalar invariants as $(j = 1, 2, ...n; k = 2, 3..., m)$

$$r_j = R^{a_1}_{a_2} R^{a_2}_{a_2} ... R^{a_n}_{a_1} \qquad c_k = C^{a_1 b_1}{}_{a_2 b_2} C^{a_2 b_2}{}_{a_3 b_3} ... C^{a_k b_k}{}_{a_1 b_1} \tag{10}$$

and we name $\{\beta^j\}$ the maximal set of functionally independent scalar invariants. It follows from equation (9) that

$$X r_j = X c_k = 0 \tag{11}$$

for every $j = 1 \ldots n$, $k = 1 \ldots m$

We name the number of linearly independent scalar invariants d which is equal to the rank of the system

$$rank\,\{dr_1 ... dr_n; dc_1 ... dc_m\} = d \qquad d \leq n$$

. We note: if $d = n$ the solution for the Killing vector field is trivial, $X = 0$. However if $d < n$ the space time could admit Killing vectors. We define $\{\beta^j\}$ as the maximal set of functionally independent scalar invariants and find that any Killing vector satisfies

$$i_X (d\beta^1 \wedge d\beta^2 \wedge ... \wedge d\beta^d) = 0. \tag{12}$$

A Killing vector is called smooth if it is unlimited differentiable, i. e. of class $\mathcal{C}^\infty$. Smooth Killing vectors form a Lie algebra, because it satisfies

$$\pounds_{[X,Y]} T = \pounds_X (\pounds_Y T) - \pounds_Y (\pounds_X T) \tag{13}$$

This itaration of the computation of the Lie brackets works unlimited, except for the case that X or Y are not differentiable.

If T_{ab} is of class $\mathcal{C}^\infty$, then the solution X^a of equation (3) is $\mathcal{C}^\infty$ as well.

3 Plane symmetric space time

We will examine the Ricci collineations for the general static plane symmetric space time, taken from [9]

$$ds^2 = -e^{2\alpha(x)} dt^2 + dx^2 + e^{2\beta(x)} (dy^2 + dz^2) \tag{14}$$

were we work in the signature (-,+,+,+).
Plane symmetric space times are considered interesting since they possess a physical Energy-Momentum tensor [10]. The space time we discuss here is static since the functions of the metric do not depend on time, just on the x coordinate.

The metric (14) has a diagonal Ricci tensor which also just depends on x. The Ricci tensor is obtained using *mathematica* with the package *diffgeo*.

$$R_{ab} = \begin{pmatrix} f_1(x) & 0 & 0 & 0 \\ 0 & f_2(x) & 0 & 0 \\ 0 & 0 & f_3(x) & 0 \\ 0 & 0 & 0 & f_3(x) \end{pmatrix} \tag{15}$$

with the functions

$$\begin{aligned} f_1 &= e^{2\alpha(x)}(\alpha'(x)^2 + 2\alpha'(x)\beta'(x) + \alpha''(x)) \\ f_2 &= -(\alpha'(x)^2 + \alpha''(x) + 2(\beta'(x)^2 + \beta''(x))) \\ f_3 &= -e^{2\beta(x)}(\alpha'(x)\beta'(x) + 2\beta'(x)^2 + \beta''(x)) \end{aligned} \tag{16}$$

We will call this tensor the symmetric tensor T_{ab} and examine its collineation fields depending on its rank or grade of degeneracy, by setting some of its components, i.e. by setting the functions that correspond to the component, adequately to 0. The new tensor with a smaller dimension will be written as $T_{\mu\nu}$.
Small Latin letters $\{a, b..\}$ demonstrate that we work with all 4 coordinates and that the tensor has the full rank 4, while Greek letters $\{\mu, \nu...\}$ indicate that we just take into account the coordinates that are non-zero in T_{ab}. Big Latin letters $\{A, B..\}$ correspond to the coordinates that are 0 in the case that is analyzed.

3.1 Rank 4: $f_1, f_2, f_3 \neq 0$

If T_{ab} has the full rank $rank(T) = 4$ we can take this tensor as a non-degenerate metric tensor and equation (1) for the Ricci collineations can be seen as a usual Killing equation.

Taking T_{ab} (eq.15) as a metric tensor we obtain the following non-vanishing Christoffel symbols

$$\begin{gathered} \Gamma^x_{tt} = -\frac{f_1'}{2f_2} \qquad \Gamma^x_{xx} = \frac{f_2'}{2f_2} \qquad \Gamma^x_{yy} = \Gamma^x_{zz} = -\frac{f_3'}{2f_2} \\ \Gamma^t_{xt} = \frac{f_1'}{2f_1} \qquad \Gamma^y_{xy} = \Gamma^z_{xz} = \frac{f_3'}{2f_3} \end{gathered} \tag{17}$$

as well as the non-vanishing components of the Riemann tensor and the Ricci Tensor and the Ricci scalar. The computation was done with the help of *mathematica* package *diffgeo*.

$$\begin{gathered} R_{ytyt} = R_{ztzt} = -\frac{f_1' f_3'}{4f_2} \qquad R_{txtx} = \frac{1}{4}\left(\frac{f_1'^2}{f_1} + \frac{f_1' f_2'}{f_2} + 2f_1''\right) \\ R_{xyxy} = R_{xzxz} = \frac{1}{4}\left(\frac{f_2' f_3'}{f_2} - \frac{f_3'^2}{f_3} - 2f_3''\right) \qquad R_{yzyz} = \frac{f_3'^2}{4f_2} \end{gathered} \tag{18}$$

$$\begin{aligned} R_{xx} &= \frac{f_1 f_3 f_2'(f_3 f_1' + 2f_1 f_3') + f_2(2f_1^2 f_3'^2 + f_3^2(f_1'^2 - 2f_1 f_1'') - 4f_1^2 f_3 f_3''))}{4f_1^2 f_2 f_3^2} \\ R_{tt} &= \frac{f_1 f_3 f_1' f_2' + f_2(-2f_1 f_1' f_3' + f_3(f_1'^2 - 2f_1 f_1''))}{4f_1 f_2^2 f_3} \\ R_{yy} = R_{zz} &= \frac{f_1 f_2' f_3' - f_2(f_1' f_3' + 2f_1 f_3'')}{4f_1 f_2^2} \end{aligned} \tag{19}$$

Futhermore we calculated the non-vanishing components of the Weyl tensor

$$\begin{aligned} &W^{yt}{}_{yt} = W^{zt}{}_{zt} = W^{xy}{}_{xy} = W^{xz}{}_{xz} = h(x) \qquad \text{and} \qquad W^{tx}{}_{tx} = W^{yz}{}_{yz} = 2h(x) \qquad \text{with} \\ &h(x) = \frac{f_1 f_3 f_2'(-f_3 f_1' + f_1 f_3') - f_2(-2f_1^2 f_3'^2 + f_3^2(f_1'^2 - 2f_1 f_1'') + f_1 f_3(f_1' f_3' + 2f_1 f_3''))}{24 f_1^2 f_2^2 f_3^2} \end{aligned} \tag{20}$$

To solve the Killing equation (3) we proceed as explained in section (2). Looking at the Riemann tensor (19), Ricci tensor (17) and Weyl tensor (20) we obtain certain constrains on the 10 searched parameters $X_a, \Omega_{\mu\nu}$.

First of all we find that the scalar invariants (10) can just depend on x. Hence there is just one functionally independent scalar invariant, $rank\{dr_1 ... dr_4; dc_1 ... dc_6\} = 1$ with $dr_j = r'dx; dc_l = c'dx$. As above we name it β^1 and find

$$d\beta^1 \propto dx \qquad X^\mu \partial_\mu \beta^1 = 0 \qquad \Leftrightarrow \qquad X^x = 0 \tag{21}$$

which is the first constrain on the Killing vector field X.

We now analyze the part of the Ricci tensor of equation (9). Equation (17) shows that $R_{ab} = 0$ for $a \neq b$ and $R_{aa} = R_{aa}(x)$. We find

$$\pounds_X R_{ab} = X^e \partial_e R_{ab} + 2R_{e(b}\partial_{a)} X^e = X^x \partial_x R_{ab}(x) + 2R_{e(b}\partial_{a)} X^e = R_{bb} \partial_a X^b + R_{aa} \partial_b X^a \tag{22}$$

where eq.(21) has been included. We can specify the following cases

- If $\{a = b = x\}$ the equation is always fulfilled.
- If $\{a = x; b = j \neq x\}$

$$R_{jj} \partial_x X^j = 0 \tag{23}$$

Knowing that the Ricci tensor does not vanish in all diagonal components R_{jj} we obtain that for all j.

$$\partial_x X^j = 0. \tag{24}$$

- If $\{a = j \neq x; b = l \neq x\}$

$$R_{ll} \partial_j X^l + R_{jj} \partial_l X^j = 0 \tag{25}$$

Identify the skewsymmetric tensor Ω_{ab} as

$$\nabla_a X_b = \Omega_{ab} \tag{26}$$

we obtain that for the latter case $\{a = j \neq x; b = l \neq x\}$ the covariant derivative is just a partial one:

$$\partial_j X^l = \nabla_j X^l - \Gamma^l_{ja} X^a = \Omega_j{}^l \tag{27}$$

because just for $a = x$ the Christoffel symbol does not vanish, but in this case $X^a = X^x = 0$. From this and equation (25) we get that for the case $\{a = j \neq x; b = l \neq x\}$

$$(R^l_l - R^j_j)\Omega_{lj} = 0 \tag{28}$$

which means that either $R^l_l = R^j_j$ or $\Omega_{lj} = 0$. In the generic case $R^t_t \neq R^y_y = R^z_z$ we have that

$$\Omega_{ty} = \Omega_{tz} = 0 \tag{29}$$

and Ω_{yz} is not determined.
From this and equation (24) we see that $\partial_a X^t = 0$. Hence X^t can just depend on a constant which we will call $\tilde{c}$.

Having explored the beneficing of the Ricci tensor we proceed due to eq. (9) analogously with the Weyl tensor. However, this just leads to the same conditions that we already got from the Ricci tensor and does not give us new information about the system.

We continue with the actual Killing equation (3 on page 4), but this time written in terms of the Lie derivative.

$$X^t = \tilde{c} \tag{30}$$

$$\pounds_X T_{ab} = X^d \partial_d T_{ab} + \partial_a X^d T_{db} + \partial_b X^d T_{ad} \tag{31}$$

Using all the information we already obtained about X we see that the first term cancels completely and for $j, k \neq x$ the other two reduce to

$$f_k(x) \partial_j X^k + f_j(x) \partial_k X^j = 0 \tag{32}$$

where the index of f_μ has no summation, but marks whether we mean f_1 for $\mu = t$ or f_3 for $\mu = y, z$. In eq. (32) as in the following equation (33) $j, k \neq x$. Contracting $T_{\mu\nu}$ and X^ν which we can do because the partial derivative for $\mu \neq x$ commutes with the metric, (36) leads to

$$\partial_j X_k + \partial_k X_j = 0 \tag{33}$$

This is the Killing equation in 3 dimensional flat space. Writing it in terms of the skewsymmetring tensor $\partial_a X_b = \Omega_{ab}$ we have under the integrability condition $\partial_l \Omega_{jk} = 0$ we see that

$$\Omega_{yz} = \Omega_{yz}(x). \tag{34}$$

As a matter of fact, (23) implies that $\partial_x \partial_i X^j = 0$ and therefore $\partial_x \Omega_{ij} = 0$ which includes (28) and that $\Omega_{yz} = -\Omega_{zy} = c, c =$constant. This leads to the solution

$$\Omega_{yz} = c f_3 \qquad\qquad c = constant \tag{35}$$

For X_y, X_z we get from equations (29), (24) and from the Killing equation itself that the most general form is

$$X_y = f_3(c_1 - cz) \qquad\qquad X_z = f_3(c_2 + cy) \tag{36}$$

which adds up to the most general Killing vector field, with all the c being constants

$$X^a = \begin{pmatrix} \tilde{c} \\ 0 \\ (c_1 - cz) \\ (c_2 + cy) \end{pmatrix} \tag{37}$$

$$\text{or} \qquad X = \tilde{c}\, \partial_t + c_1\, \partial_y + c_2\, \partial_z + c\,(y \partial_z - z \partial_y)$$

This is a linear combination of the independent infinitesimal generators or Killing vector fields. The number of independent constants is equal to the number of number of independent symmetries. We can construct their infinitesimal generators by setting one constant of the general solution X^a to 1 and the others to 0. We arrive at 4 independent Killing vector fields

$$X^1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \qquad X^2 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \qquad X^3 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \qquad X^4 = \begin{pmatrix} 0 \\ 0 \\ -z \\ y \end{pmatrix}. \tag{38}$$

This number is also equal to the dimension of the Lie algebra, so we see that the dimension of the Lie algebra of the Killing vectors is $dim = 4$.

3.2 Rank 3

For the rank of $T_{\mu\nu}$ to be 3 there are two possibilities, either (A) $f_1 = 0$ and $f_2, f_3 \neq 0$ or (B) $f_2 = 0$ and $f_1, f_3 \neq 0$. In case A, x is a coordinate on the hypersurface $x^4 = t$, constant; and the metric coefficients $T_{\mu\nu}(x)$ are not constant, but the metric $T_{\mu\nu}$ does not change in passing from one sheet to another.

Contrarily, in case B $x^4 = x$, which is constant on every sheet, $T_{\mu\nu}$ are constant too and the Christoffel symbols and curvature vanish on the hypersurface $x^4 = x$, constant. However the metric varies from one sheet to another.

We will discuss a piece of theory and later on discuss both cases separately:
For $rank\,\{T_{\mu\nu}\} = 3$ we can write the collineation field X that satisfies the condition (1) $\pounds_x T = 0$ as

$$X = Z^\alpha \partial_\alpha + \phi \partial_4 \tag{39}$$

where coordinate 4 is the one that does not appear in $T_{\mu\nu}$, i. e. $T_{4a} = 0$. In components we get

$$\pounds_X T_{ab} = \pounds_Z T_{ab} + \phi \partial_4 T_{ab} = 0 \tag{40}$$

which translates to

$$\partial_4 Z^\alpha = 0 \qquad \nabla_{(\alpha} Z_{\beta)} + \phi K_{\alpha\beta} = 0 \qquad \text{with} \qquad K_{\alpha\beta} := \frac{1}{2}\partial_4 T_{\mu\nu}\,. \tag{41}$$

This only detemines the symmetric part of $\nabla_\alpha Z_\beta$ and the skewsymmtric part is arbitrary, hence

$$\nabla_\alpha Z_\beta = \Omega_{\alpha\beta} - \phi K_{\alpha\beta} \qquad \Omega_{\alpha\beta} + \Omega_{\beta\alpha} = 0 \tag{42}$$

with $\Omega_{\alpha\beta}$ being a skewsymmetric tensor.

The integrability conditions for (42) together with the Jacobi identity for $\nabla_\alpha \Omega_{\beta\mu}$ lead to

$$\nabla_\mu \Omega_{\alpha\beta} = Z^\rho R_{\rho\mu\alpha\beta} + \nabla_\beta(\phi K_{\mu\alpha}) - \nabla_\alpha(\phi K_{\mu\beta}) \tag{43}$$

Further integrability conditions coming from the discussion of the Killing equation (43) translate to

$$\pounds_Z R_{\mu\nu} = \phi(K^{\alpha\beta} R_{\mu\alpha\nu\beta} - R^\alpha_{(\mu} K_{\nu)\alpha}) + \nabla^\alpha \nabla_\alpha(\phi K_{\mu\nu}) + \nabla^\mu \nabla_\nu(\phi K^\alpha_\alpha) \tag{44}$$

From the commutation relations of ∇_ν and ∂_4 on the equations (41), (42) we get

$$\partial_4 \Omega_{\alpha\beta} = 2\Omega_{\lambda[\beta} K^{\lambda}_{\alpha]} + 2Z^{\lambda} \nabla_{[\alpha} K_{\beta]\lambda} \tag{45}$$

for the antisymmetric part and

$$\pounds_Z K^{\mu}_{\beta} + \dot{\phi} K^{\mu}_{\beta} + \phi \dot{K}^{\mu}_{\beta} = 0 \tag{46}$$

for the symmetric part where the dot marks the derivative with respect to coordinate x^4.

3.2.1 Case A: $f_1 = 0 \qquad f_2 \neq 0$ and $f_3 \neq 0$

Starting with the case $f_1 = 0$ the space time geometry reduces to a 3 dimensional hypersurface with the t coordinate being constant. The collineation field X satisfies the condition $\pounds_X T = 0$. This manifold has curvature. We calculate the non-vanishing Christoffel symbols, the Riemann and the Ricci tensor of $T_{\mu\nu}$.

$$\Gamma^x_{xx} = \frac{f_2'}{2f_2} \qquad \Gamma^x_{yy} = \Gamma^x_{zz=} - \frac{f_3'}{2f_2} \qquad \Gamma^y_{xy} = \Gamma^z_{xz} = \frac{f_3'}{2f_3} \tag{47}$$

$$R_{yzyz} = \frac{-f_3'^2}{4f_2} \qquad R_{xyxy} = R_{xzxz} = \frac{1}{4}(\frac{f_2' f_3'}{f_2} + \frac{f_3'^2}{f_3} - 2f_3'') \tag{48}$$

$$R_{xx} = \frac{f_3 f_2' f_3' + f_2 f_3'^2 - 2f_2 f_3 f_3''}{2f_2 f_3^2} \qquad R_{yy} = R_{zz} = \frac{f_2' f_3' - 2f_2 f_3 f_3''}{4f_2^2} \tag{49}$$

Our tensors are

$$T_{\mu\nu} = diag(f_2(x), f_3(x), f_3(x)) \qquad K_{\alpha\beta} = \frac{1}{2}\partial_t T_{\alpha\beta} = 0 \qquad \text{with} \qquad x^4 = t. \tag{50}$$

Hence the equation (40) $\pounds_X T_{ab} = 0$ reduces to $\pounds_Z T_{\alpha\beta} = 0$ which is a usual Killing equation in 3 dimensions.

$$\nabla_{(\alpha} Z_{\beta)} = 0 \tag{51}$$

with no further condition on ϕ.

We proceed similar to the case of $rank(T) = 4$ but, as in three dimensions the Riemann tensor has as many independent components as the Ricci tensor, the integrability conditions will be simpler.

Again as in section (3.1) there is at most one functionally independent scalar invariant, $rank\,\{dr_1 ... dr_3\} = 1$ with $dr_j = r' dx$. As above we name it β^1 and find

$$\beta^1 \propto dx \qquad Z^{\mu}\partial_{\mu}\beta^1 = 0 \qquad \Leftrightarrow \qquad Z^x = 0. \tag{52}$$

This is the same case as in section (3.1) if we forget about time, with equation (33) we can directly write Z.

$$Z_{\mu} = \begin{pmatrix} 0 \\ f_3(c_1 - cz) \\ f_3(c_2 + cy) \end{pmatrix} \qquad Z^{\mu} = \begin{pmatrix} 0 \\ (c_1 - cz) \\ (c_2 + cy) \end{pmatrix} \tag{53}$$

with all the c being constants.

To find the collineations for the rank 3 tensor

$$T_{ab} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & f_2 & 0 & 0 \\ 0 & 0 & f_3 & 0 \\ 0 & 0 & 0 & f_3 \end{pmatrix} \tag{54}$$

one just has to add the part of the t coordinate to gain the collineation field

$$X = Z^\alpha \partial_\alpha + \phi \partial_t \tag{55}$$

with ϕ being an arbitrary function.

$$X^a = \begin{pmatrix} \phi \\ 0 \\ (c_1 - cz) \\ (c_2 + cy) \end{pmatrix} \tag{56}$$

$$\text{or} \qquad X = \phi\, \partial_t + c_1 \partial_y + c_2 \partial_z + c_3(y\partial_z - z\partial_y)$$

The infinetesimal generators are

$$X^1 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \qquad X^2 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \qquad X^3 = \begin{pmatrix} 0 \\ 0 \\ -z \\ y \end{pmatrix}. \tag{57}$$

However, since ϕ is arbitrary they can be an infinite number of other collineation vectors X^i, $i = 4...\infty$ and the Lie algebra of the collineation vectors is infinite dimensional.

3.2.2 Case B: $f_2 = 0 \qquad f_1 \neq 0$ and $f_3 \neq 0$

Space time is foliated in sheets labeled by $x^4 = x$, constant.

The tensor $T_{\mu\nu}$ depends on x and is different in each sheet, but does not depend on the sheet's coordinates, hence the Christoffel symbols and the curvature tensor vanish. The covariant derivative reduces to a partial one. Our tensors become

$$T_{\mu\nu} = \begin{pmatrix} f_1 & 0 & 0 \\ 0 & f_3 & 0 \\ 0 & 0 & f_3 \end{pmatrix} \qquad K_{\alpha\beta} = \frac{1}{2} \begin{pmatrix} f_1' & 0 & 0 \\ 0 & f_3' & 0 \\ 0 & 0 & f_3' \end{pmatrix}. \tag{58}$$

We use the equations (39) to (46) to give conditions on ϕ and Z which leads to the most general collineation field X.

Procedure

Due to the vanishing of the Riemann tensor equation (44) reduces to

$$K_{\mu\nu} T^{\alpha\beta} \phi_{\alpha\beta} + K^\alpha_\alpha \phi_{\mu\nu} - 2K^\alpha_{(\nu}\phi_{\mu)\alpha} = 0 \tag{59}$$

we gain a system of linear equations constraining the second derivatives of ϕ where ϕ_i means $\partial_i \phi$. We distinguish $\mu = \nu$:

for $\mu = \nu$

$$\begin{pmatrix} \frac{2f_3'}{f_3} & \frac{f_1'}{f_3} & \frac{f_1'}{f_3} \\ \frac{f_3'}{f_1} & \frac{f_1'}{f_1} + \frac{f_3'}{f_3} & \frac{f_3'}{f_3} \\ \frac{f_3'}{f_1} & \frac{f_3'}{f_3} & \frac{f_1'}{f_1} + \frac{f_3'}{f_3} \end{pmatrix} \begin{pmatrix} \phi_{tt} \\ \phi_{yy} \\ \phi_{zz} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \tag{60}$$

This is a homogeneous linear system on the second derivatives of ϕ with the determinant

$$Det = \frac{4f_1' f_3'^2}{f_1 f_3^2}.$$

for $\mu \neq \nu$ eq. (59) reads

$$\begin{aligned} f_3'\phi_{ty} = 0 \quad &\Leftrightarrow \quad \left[f_3' = 0 \quad or \quad \phi_{ty} = 0\right] \\ f_3'\phi_{tz} = 0 \quad &\Leftrightarrow \quad \left[f_3' = 0 \quad or \quad \phi_{tz} = 0\right] \\ f_1'\phi_{yz} = 0 \quad &\Leftrightarrow \quad \left[f_1' = 0 \quad or \quad \phi_{yz} = 0\right] \end{aligned} \tag{61}$$

The cases $f_1' = 0, f_3' = 0$ will be discussed later on.

$$0 = \partial_\alpha Z^\rho K_{\rho\beta} + \partial_\beta Z^\rho K_{\alpha\rho} + \phi \partial_x K_{\alpha\beta} + \partial_x \phi K_{\alpha\beta} \tag{62}$$

were we used that $Z^\rho \partial_\rho K_{\alpha\beta} = 0$. This gives constrains on the x-dependence of ϕ.

$$\partial_\alpha Z^\rho = \partial_\alpha (T^{\gamma\rho} Z_\gamma) = T^{\gamma\rho}(\partial_\alpha Z_\gamma) = T^{\gamma\rho}(\Omega_{\alpha\gamma} - \phi K_{\alpha\gamma}) \tag{63}$$

were the ∂_α and $T^{\gamma\rho}$ commute because the symmetric tensor just depend on the coordinate x that is is not a coordinate on the hypersurface.

for $\alpha = \beta$ eq. (59) reads

$$\partial_x \left(\frac{f_j'}{f_j} \phi \right) = 0 \qquad \{j = 1, 3\} \tag{64}$$

And $\alpha \neq \beta$ leads to

$$\frac{f_3'}{f_3}\Omega_{ty} + \frac{f_1'}{f_1}\Omega_{yt} = 0 \qquad \frac{f_3'}{f_3}\Omega_{tz} + \frac{f_1'}{f_1}\Omega_{zt} = 0 \tag{65}$$

From equation (45) we get

$$\partial_x \Omega_{\alpha\gamma} = \Omega_{\rho\gamma} K^\rho_\alpha - \Omega_{\rho\alpha} K^\rho_\gamma \tag{66}$$

the x-dependence of $\Omega_{\alpha\gamma}$ can be factored out. If we perform a change of coordinates $\Omega_{\alpha\gamma} = t_{\alpha\mu} t_{\gamma\nu} W^{\mu\nu}$ and the ansatz $t_{\alpha\mu} = diag(\sqrt{f_1}, \sqrt{f_3}, \sqrt{f_3})$ we calculate:

$$\partial_x t_{\alpha\mu} = t'_{\alpha\mu} = diag\left(\frac{f_1'}{2\sqrt{f_1}}, \frac{f_3'}{2\sqrt{f_3}}, \frac{f_3'}{2\sqrt{f_3}}\right) = K_{\alpha\rho} T^{\rho\sigma} t_{\sigma\mu} \tag{67}$$

and this substituted into eq. (66) leads to $\qquad t_{\alpha\mu} t_{\gamma\nu} W'^{\mu\nu} = 0$

Hence we see that $W^{\mu\nu}$ does not depend on x.

$$\partial_x W^{\mu\nu} = 0 \qquad\qquad W^{\mu\nu} = W^{\mu\nu}(t, y, z) \tag{68}$$

The general solution of eq: (66) is thus

$$\Omega_{\mu\nu} = \begin{pmatrix} 0 & \sqrt{f_1}\sqrt{f_3}W^{ty} & \sqrt{f_1}\sqrt{f_3}W^{tz} \\ -\sqrt{f_1}\sqrt{f_3}W^{ty} & 0 & f_3W^{yz} \\ -\sqrt{f_1}\sqrt{f_3}W^{tz} & -f_3W^{yz} & 0 \end{pmatrix} \tag{69}$$

Through the following equation obtained from equation (43) the t, y and z dependence of the $\Omega_{\alpha\gamma}$ can be estimated

$$\partial_\mu \Omega_{\alpha\gamma} = K_{\mu[\alpha}\phi_{\gamma]} = K_{\mu\alpha}\phi_\gamma - K_{\mu\gamma}\phi_\alpha. \tag{70}$$

Putting in components leads to the conditions on $W^{\mu\nu}$

$$\begin{array}{ccc} \partial_t W^{ty} = \frac{f_1'}{2\sqrt{f_1}\sqrt{f_3}}\phi_y & \partial_y W^{ty} = -\frac{f_3'}{2\sqrt{f_1}\sqrt{f_3}}\phi_t & \partial_z W^{ty} = 0 \\ \partial_t W^{tz} = \frac{f_1'}{2\sqrt{f_1}\sqrt{f_3}}\phi_z & \partial_y W^{tz} = 0 & \partial_z W^{tz} = -\frac{f_3'}{2\sqrt{f_1}\sqrt{f_3}}\phi_t \\ \partial_t W^{yz} = 0 & \partial_y W^{yz} = \frac{f_3'}{2f_3}\phi_z & \partial_z W^{yz} = -\frac{f_3'}{2f_3}\phi_y \end{array} \tag{71}$$

The t, y and z dependence of the Z_γ is given by

$$\partial_\alpha Z_\gamma = \Omega_{\alpha\gamma} - \phi K_{\alpha\gamma} \tag{72}$$

which gives the constrains

$$\begin{array}{ccc} \partial_t Z_t = -\phi\frac{1}{2}f_1' & \partial_y Z_y = -\phi\frac{1}{2}f_3' & \partial_z Z_z = -\phi\frac{1}{2}f_3' \\ \partial_y Z_t = -\Omega_{ty} & \partial_t Z_y = \Omega_{ty} & \partial_t Z_z = \Omega_{tz} \\ \partial_z Z_t = -\Omega_{tz} & \partial_z Z_y = -\Omega_{yz} & \partial_y Z_z = \Omega_{yz} \end{array} \tag{73}$$

Four different kinds of solutions can be distinguished specifying the rank of the system.

Case $f_1'(x) = 0$ $f_3'(x) \neq 0$

Putting f_1' to 0 reduces the system of equations.

Eq. (60) shows that

$$\phi_{yy} = -\phi_{zz} \tag{74}$$

which means that ϕ is a harmonic function $\Delta^2\phi = 0$. Eq. (61) leads to

$$\phi_{tt} = \phi_{ty} = \phi_{tz} = 0. \tag{75}$$

Eq. (64) gives the condition that ϕ needs to have an overall factor of $\frac{f_3}{f_3'}$.

$$\partial_x(\frac{f_3'}{f_3}\phi) = 0 \qquad \Leftrightarrow \qquad \phi = \frac{f_3}{f_3'}\psi(t, y, z) \tag{76}$$

Equation (71) shows that $\partial_t W^{ty} = \partial_z W^{ty} = 0$; $W^{ty} = W^{ty}(y)$ and

$$\partial_y W^{ty} = -\frac{f_3'}{2\sqrt{f_1}\sqrt{f_3}}\phi_t = -\frac{\sqrt{f_3(x)}}{2\sqrt{f_1}}\partial_t\psi(t, y, z) \tag{77}$$

But W^{ty} has no x dependence, hence this statement is inconsistent with $f_3' \neq 0$ unless

$$\partial_t\psi(t, y, z) = 0 \qquad \text{and} \qquad \partial_y W^{ty} = 0 \tag{78}$$

which means that $W^{ty} = constant$.

Rewriting eq. (71) in terms of ψ

$$\partial_t W^{yz} = 0 \qquad \partial_y W^{yz} = \frac{1}{2}\partial_z \psi \qquad \partial_z W^{yz} = -\frac{1}{2}\partial_y \psi \tag{79}$$

From the two latter equations we see that W^{yz} and $\frac{1}{2}\psi$ satisfy together as a pair of functions the Cauchy-Riemann conditions [11]. Hence

$$W^{yz} + \frac{i}{2}\psi = m(y + iz) \tag{80}$$

with m being an analytic function of the complex variable $\omega = y + iz$.

and

$$Z^t = c \qquad \text{c=constant and} \qquad Z^y = Z^y(y,z) \qquad \text{c=constant and} \qquad Z^z = Z^z(y,z) \tag{81}$$

Introducing the complex function $M(y + iz) = Z^y(y,z) + iZ^z(y,z)$ and including eq. (73) and (80) we can write

$$\frac{dM}{d\omega} = im(\omega). \tag{82}$$

This leads to

$$Z^y = \mathbb{R}eM(y+iz) \qquad Z^z = \mathbb{I}mM(y+iz) \qquad \psi = 2\mathbb{I}mf = -2\mathbb{R}eM' \tag{83}$$

This lead to the collineation field

$$X^a = \begin{pmatrix} c \\ \phi \\ \mathbb{R}eM(y+iz) \\ \mathbb{I}mM(y+iz) \end{pmatrix} \qquad \phi = -2\frac{f_3(x)}{f_3'(x)}\mathbb{R}eM'(y+iz) \tag{84}$$

where $M(\omega)$ is an arbitrary analytic function. Since M is analytic, $\mathbb{R}eM'$ is a harmonic function [11], in accordance with eq. (75). The collineation field X depends on the arbitrary function $M(y + iz)$ and the algebra of collineation fields is infinite dimensional.

Case $f_1'(x) \neq 0 \; f_3'(x) = 0$

Analogously to the previos case, eq. (64) gives the condition that this time ϕ needs to have an overall factor of $\frac{f_1}{f_1'}$.

$$\partial_x(\frac{f_1'}{f_1}\phi) = 0 \qquad \Leftrightarrow \qquad \phi = \frac{f_1}{f_1'}\psi(t,y,z) \tag{85}$$

Again we follow equation (71): $\partial_y W^{ty} = \partial_z W^{ty} = 0$; $W^{ty} = W^{ty}(t)$ and

$$\partial_t W^{ty} = -\frac{f_1'}{2\sqrt{f_1}\sqrt{f_3}}\phi_y = -\frac{\sqrt{f_1(x)}}{2\sqrt{f_3}}\partial_y\psi(t,y,z) \tag{86}$$

But W^{ty} has no x dependence, hence this statement is inconsistent with $f_1' \neq 0$ unless

$$\partial_y\psi(t,y,z) = 0 \qquad \text{and} \qquad \partial_t W^{ty} = 0 \tag{87}$$

which means that W^{ty} is a constant c_1 and $\Omega_{ty} = \sqrt{f_1}\sqrt{f_3}c_1$, where we remind the f_3 is also constant. In the same matter, $\partial_y W^{tz} = \partial_z W^{tz} = 0$; $W^{tz} = W^{tz}(t)$.

$$\partial_t W^{tz} = -\frac{f_1'}{2\sqrt{f_1}\sqrt{f_3}}\phi_z = -\frac{\sqrt{f_1(x)}}{2\sqrt{f_3}}\partial_z\psi(t,y,z) \tag{88}$$

Again, W^{tz} has no x dependence, hence this statement is inconsistent with $f_1' \neq 0$ unless

$$\partial_z\psi(t,y,z) = 0 \qquad \text{and} \qquad \partial_t W^{tz} = 0 \tag{89}$$

So also W^{tz} is a constant c_2 and $\Omega_{tz} = \sqrt{f_1}\sqrt{f_3}c_2$ and

$$\psi = \psi(t) \qquad \Leftrightarrow \qquad \phi = \frac{f_1}{f_1'}\psi(t). \tag{90}$$

Looking at the third of the 3 equations of (71) we find that

$$\partial_\mu W^{yz} = 0 \qquad \Leftrightarrow \qquad W^{yz} = c_3 \qquad c_3 = \text{constant} \qquad \Omega_{yz} = f_3 c_3 \tag{91}$$

Applying equation (60) we find that there are no constrains on ϕ_{tt}. Eq. (61) gives $\phi_{yz} = 0$. Translating to eq. (72) we see

Z_z

$$\partial_z Z_z = 0 \qquad \partial_y Z_z = f_3 c_3 \qquad \partial_t Z_z = \sqrt{f_1}\sqrt{f_3}c_1 \tag{92}$$

But f_1 depends on x and $\partial_x Z^\mu = 0$ which is a contradiction unless

$$\partial_t Z_z = 0 \qquad c_1 = 0. \tag{93}$$

Z_y

$$\partial_y Z_y = 0 \qquad \partial_z Z_y = -f_3 c_3 \qquad \partial_t Z_y = -\sqrt{f_1}\sqrt{f_3}c_2 \tag{94}$$

Again f_1 depends on x and $\partial_x Z^\mu = 0$ which is a contradiction unless

$$\partial_t Z_y = 0 \qquad c_2 = 0. \tag{95}$$

Z_t

$$\partial_t Z_t = -\frac{1}{2}f_1\psi(t) \qquad \partial_y Z_t = 0 \qquad \partial_z Z_t = 0 \tag{96}$$

With this conditions we can write Z^μ and the collineation field X^a

$$Z^\mu = \begin{pmatrix} -\frac{1}{2}\Psi(t) \\ -(c_3 z + k_2) \\ (c_3 y + k_1) \end{pmatrix} \qquad X^a = \begin{pmatrix} -\frac{1}{2}\Psi(t) \\ \phi \\ -(c_3 z + k_2) \\ (c_3 y + k_1) \end{pmatrix} \tag{97}$$

or

$$X = -\frac{1}{2}\Psi(t)\,\partial_t + \frac{f_1(x)}{f_1'(x)}\,\Psi'(t)\partial_x + k_2\partial_y + k_3\partial_z + c_3(y\partial_z - z\partial_y)$$

with $\Psi(t) = \int dt\psi(t)$, $\phi = \frac{f_1}{f_1'}\psi(t)$ and $\psi(t)$ being an arbitrary function, just depending on t. From here we get the infinitesimal generators

$$X^1 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \qquad X^2 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \qquad X^3 = \begin{pmatrix} 0 \\ 0 \\ -z \\ y \end{pmatrix} \tag{98}$$

and an infinite number of collineation vectors depending on the arbitrary function ψ. Hence, also the dimension of the Lie algebra is infinite.

Case $f_1'(x) \neq 0$ $f_3'(x) \neq 0$

Eq. (64) gives the condition that this time ϕ needs to have an overall factor of $\frac{f_1}{f_1'}$, but also of $\frac{f_3}{f_3'}$.

$$\partial_x(\frac{f_j'}{f_j}\phi) = 0 \qquad \Leftrightarrow \qquad \phi = \frac{f_j}{f_j'}\psi(t,y,z) \qquad j = 1,3 \tag{99}$$

we get the condition that this time ϕ needs to have an overall factor of $\frac{f_1}{f_1'} = \frac{f_3}{f_3'}$.

or ϕ is simply 0. We will discuss the latter case at first

• $\phi = 0$

In this case we have a usual Killing equation in 3 dimensions.

• $\frac{f_1'}{f_1} = \frac{f_3'}{f_3} = \tilde{f}$ where $\tilde{f}$ is a constant.

In this case we can write, due to eq. (71)

$$\phi = \frac{f_j}{f_j'}\psi(t,y,z). \tag{100}$$

Eq. (61) shows that all the mixed second derivatives of ϕ are 0. The system of equations (60) has non-vanishing determinant, hence $\phi_{tt} = \phi_{yy} = \phi_{zz} = 0$,

$$\frac{\sqrt{f_3'}}{\sqrt{f_1}}\phi_{tt} = -\frac{f_j'}{f_j}(2\phi_{yy} + \phi_{zz}) = -\frac{f_j'}{f_j}(\phi_{yy} + 2\phi_{zz}) = -\frac{1}{2}\frac{f_j'}{f_j}(\phi_{yy} + \phi_{zz}) \tag{101}$$

which shows that in general all second derivatives of ϕ are 0

$$\phi_{\mu\nu} = 0 \tag{102}$$

This means that ψ has to be linear in t, y, z. Eq. (71) gives

$$\begin{array}{ccc} \partial_t W^{ty} = \frac{\sqrt{f_1}}{2\sqrt{f_3}}\partial_y\psi & \partial_y W^{ty} = -\frac{\sqrt{f_3}}{2\sqrt{f_1}}\partial_t\psi & \partial_z W^{ty} = 0 \\ \partial_t W^{tz} = \frac{\sqrt{f_1}}{2\sqrt{f_3}}\partial_z\psi & \partial_y W^{tz} = 0 & \partial_z W^{tz} = -\frac{f_3'}{2\sqrt{f_1}\sqrt{f_3}}\phi_t \\ \partial_t W^{yz} = 0 & \partial_y W^{yz} = \frac{f_3'}{2f_3}\phi_z & \partial_z W^{yz} = -\frac{\sqrt{f_3}}{2\sqrt{f_1}}\partial_t\psi \end{array} \tag{103}$$

Because $W^{\mu\nu}$ can not have a x dependence we observe

$$\frac{\sqrt{f_3}}{\sqrt{f_1}} = \text{constant} \qquad \text{or} \qquad \partial_\mu\phi = 0 \tag{104}$$

With the ansatz $\psi(t,y,z) = l_1 t + l_2 y + l_3 z + l$ we get for the Ω

$$\Omega_{ty} = \frac{1}{2}f_1(l_2 t + k_1) - \frac{1}{2}f_3(l_1 y + k_2) \qquad \Omega_{tz} = \frac{1}{2}f_1(l_3 t + k_3) - \frac{1}{2}f_3(l_1 z + k_4) \qquad \Omega_{tz} = \frac{1}{2}f_3(l_3 y + l_2 z + k_3 + k) \tag{105}$$

Eq. (73) constrains ψ to $l_1 = l_2 = l_3 = 0$ so $\psi = l$ is a constant. This means $\partial_\mu \psi = 0$ and all the $W^{\mu\nu}$ are constants which we will call k_i. Hence $\frac{\sqrt{f_3}}{\sqrt{f_1}}$ does not need to be a constant.

We arrive at the collineation field

$$X^a = \begin{pmatrix} -\frac{1}{2}lt + d_1) \\ \tilde{f}l \\ -\frac{1}{2}ly + kz + d_3 \\ -\frac{1}{2}lz - ky + d_5 \end{pmatrix} \tag{106}$$

$$\text{or} \qquad X = l(\tilde{f}\partial_x - (\frac{1}{2}(\partial_t + \partial_y + \partial_z))) + k(y\partial_z - z\partial_y) + d_1\partial_t + d_2\partial_y + d_3\partial_z$$

This gives the 5 collineation vectors

$$X^1 = \begin{pmatrix} -\frac{1}{2}t \\ \tilde{f} \\ -\frac{1}{2}y \\ -\frac{1}{2}z \end{pmatrix} \qquad X^2 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \qquad X^3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \qquad X^4 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \qquad X^5 = \begin{pmatrix} 0 \\ 0 \\ -z \\ y \end{pmatrix}. \tag{107}$$

The dimension of the Lie algebra is 5.

Case $f_1'(x) = 0$ $f_3'(x) = 0$

In this case the tensor $T_{\mu\nu}$ consists of constants

$$T_{\mu\nu} = \begin{pmatrix} c_1 & 0 & 0 \\ 0 & c_3 & 0 \\ 0 & 0 & c_3 \end{pmatrix}$$

and $K_{\mu\nu}$ vanishes. All the equations that determine the dependence of ϕ on the variables vanish as well and we find that $\phi(t, x, y, z)$ is arbitrary. The Ω tensor also does not have any x-dependence, the $W_{\mu\nu}$ are constants. Following eq. (69) we rename the components $\Omega_{\mu\nu}$ as $w_{\mu\nu}$, such that they already include $\sqrt{c_1}\sqrt{c_3}$ or c_3, respectively, since they are all constants.

$$\Omega_{\mu\nu} = w_{\mu\nu} \tag{108}$$

Equation (72) reduces to the Killing equation

$$\nabla_{(\alpha} Z_{\beta)} = 0 \qquad \Leftrightarrow \qquad \partial_{(\alpha} Z_{\beta)} = 0 \tag{109}$$

due to the vanishing of the Christoffel symbols with Z being the Killing vector field on the 3 dimensional manifold tangential to manifold $x = 0$.

The collineation field becomes

$$X = Z^\alpha \partial_\alpha + \phi \partial_t \tag{110}$$

with ϕ arbitrary.

Using the constrains (73) on the Killing vector field Z in 3 dimensions we arrive at the following system of partial differential equations

$$
\begin{aligned}
\partial_t Z_t = 0 \qquad \partial_y Z_y = 0 \qquad \partial_z Z_z = 0\\
\partial_t Z_y = \Omega_{ty} = w_{ty} \qquad \partial_t Z_z = \Omega_{tz} = w_{tz} \qquad \partial_y Z_z = \Omega_{yz} = w_{yz}\\
\partial_y Z_t = -\Omega_{ty} = -w_{ty} \qquad \partial_z Z_t = -\Omega_{tz} = -w_{tz} \qquad \partial_z Z_y = -\Omega_{yz} = -w_{yz}
\end{aligned} \tag{111}
$$

It is solved by the vector field

$$
Z^\mu = \begin{pmatrix} \frac{1}{c_1}(w_{tz}z - w_{ty}y + k_1) \\ \frac{1}{c_3}(w_{ty}t - w_{yz}z + k_2) \\ \frac{1}{c_3}(w_{yz}y - w_{tz}t + k_3) \end{pmatrix} \qquad X^a = \begin{pmatrix} \frac{1}{c_1}(w_{tz}z - w_{ty}y + k_1) \\ \phi \\ \frac{1}{c_3}(w_{ty}t - w_{yz}z + k_2) \\ \frac{1}{c_3}(w_{yz}y - w_{tz}t + k_3) \end{pmatrix} \tag{112}
$$

with the k_i being constants that do not depend on $\{t, x, y, z\}$ and ϕ being arbitrary. The infinitesimal generators are

$$
\begin{aligned}
X^1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \qquad X^2 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \qquad X^3 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}\\
X^4 = \begin{pmatrix} \frac{1}{c_1}(-y) \\ 0 \\ \frac{1}{c_3}t \\ 0 \end{pmatrix} \qquad X^5 = \begin{pmatrix} \frac{1}{c_1}z \\ 0 \\ 0 \\ \frac{1}{c_3}(-t) \end{pmatrix} \qquad X^6 = \begin{pmatrix} 0 \\ 0 \\ \frac{1}{c_3}(-z) \\ \frac{1}{c_3}y \end{pmatrix}
\end{aligned} \tag{113}
$$

The dimension of the Lie algebra is infinite since there can be an infinite number of other collineation vectors depending on the constants of the arbitrary function ϕ.

3.3 Rank 2

Obtaining a T_{ab} that has $rankT = 2$ is possible if either (A) $f_1, f_2 = 0$ and $f_3 \neq 0$ or (B) $f_3 = 0$ and $f_1, f_2 \neq 0$. Due to the particular form of T_{ab}, diagonal and depending on x only, it falls in Type 2.H in the classification of reference [6] We get that equation (40) translates to

$$
\pounds_X T_{ab} = \pounds_Z T_{ab} + \phi^A \partial_A T_{ab} = 0
$$

whose different components imply that

$$
\partial_A Z^\alpha = 0 \qquad \nabla_{(\alpha} Z_{\beta)} + \phi^A K_{A|\alpha\beta} = 0 \qquad \nabla_\alpha Z_\beta = \Omega_{\alpha\beta} - \phi^A K_{A|\alpha\beta} \tag{114}
$$

with $K_{A|\alpha\beta} = \frac{1}{2}\partial_A T_{\alpha\beta}$ and $\{A = 3, 4\}$ being the coordinates that are zero in the tensor T_{ab}.

3.3.1 Case A: $f_3 \neq 0 \qquad f_2(x) = 0$ and $f_1(x) = 0$

If we have the degenerated case of the Ricci tensor with $f_1 = 0, f_2 = 0$ the new metric tensor becomes

$$
T_{\mu\nu} = \begin{pmatrix} f_3 & 0 \\ 0 & f_3 \end{pmatrix} \qquad K^t_{\alpha\beta} = \frac{1}{2}\partial_t T_{\alpha\beta} = 0 \qquad K^x_{\alpha_\beta} = \frac{1}{2}\partial_x T_{\alpha\beta} = \frac{1}{2}\begin{pmatrix} f_3' & 0 \\ 0 & f_3' \end{pmatrix}
$$

with $\alpha, \beta = y, z$ and $A, B = t, x$.

To find the collineation fields we can use similar equations as in the case were T_{ab} had rank 3 with $f_2 = 0$. Since

$$\phi^t \partial_t T_{ab} \equiv 0 \tag{115}$$

in any case the function ϕ^t is arbitrary. On the other hand, working with ϕ^x equation (59) leads to a contradiction which shows that

$$\phi^x_{yy} = \phi^x_{zz} = 0 \tag{116}$$

We distinguish two cases:

Casse A.1 $f_3' = 0$, then $K_{x|\alpha\beta} = 0$ and Z^μ is a Killing vector in 2 dimensions

$$Z^\mu = \begin{pmatrix} c_1 + cz \\ c_2 - cy \end{pmatrix} \qquad X^a = \begin{pmatrix} \phi^t \\ \phi^x \\ c_1 + cz \\ c_2 - cy \end{pmatrix} \tag{117}$$

where ϕ^t, ϕ^x are arbitraty functions of t, x, y, z.

We have the infinitesimal collineation vectors

$$X^1 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \qquad X^2 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \qquad X^3 = \begin{pmatrix} 0 \\ 0 \\ -z \\ y \end{pmatrix} \tag{118}$$

and an infinite number of collineation vectors depending on the constants of ϕ^t and ϕ^x. Hence, the dimension of the Lie algebra is infinite.

Casse A.2 $f_3' \neq 0$, we have a case that is very similar to what we discussed in section (3.2.2). Eq. (64) shows that

$$\partial_x\left(\frac{f_3'}{f_3}\phi\right) = 0 \qquad \Leftrightarrow \qquad \phi = \frac{f_3}{f_3'}\psi(t, y, z) \tag{119}$$

Rewriting eq. (71) in terms of ψ

$$\partial_y W^{yz} = \frac{1}{2}\partial_z \psi \qquad \partial_z W^{yz} = -\frac{1}{2}\partial_y \psi \tag{120}$$

From eq. (69) we have that

$$\Omega_{yz} = f_3 W^{yz} \tag{121}$$

Eq. (60) shows that

$$\phi_{yy} = -\phi_{zz} \tag{122}$$

which means that ϕ is a harmonic function $\Delta^2\phi = 0$ and so also ψ is harmonic, $\Delta^2\psi = 0$. Eq. (61) leads to

$$\phi_{yz} = 0. \tag{123}$$

This looks again like the Cauchy Riemann Conditions and we can write:

$$W^{yz} + \frac{i}{2}\psi = f(y + iz) \qquad f : \text{analytic function} \tag{124}$$

We introduce the complex function $F(y+iz) = Z^y(y,z) + iZ^z(y,z)$ and the complex variable $\omega = y + iz$:

$$\frac{dF}{d\omega} = if(\omega). \tag{125}$$

This leads to

$$Z^y = \mathbb{R}eF(y+iz) \qquad Z^z = \mathbb{I}mF(y+iz) \qquad \psi = 2\mathbb{I}mf = -2\mathbb{R}eF' \tag{126}$$

and to the collineation field

$$X^a = \begin{pmatrix} \phi^t \\ \phi^x \\ \mathbb{R}eF(y+iz) \\ \mathbb{I}mF(y+iz) \end{pmatrix} \tag{127}$$

where $F(\omega)$ is an arbitrary analytic function and ϕ^t, ϕ^x are arbitrary functions depending on t, x, y, z.

The number of collineation vectors depends on the arbitrary functions of ϕ^t, ϕ^x and $F(\omega)$. Hence, the dimension of the Lie algebra is infinite.

3.3.2 Case B: $f_1(x) \neq 0$ and $f_2(x) \neq 0$ $\quad f_3(x) = 0$

If we have the degenerated case of the Ricci tensor with $f_3 = 0$ the new metric tensor becomes

$$T_{\mu\nu} = \begin{pmatrix} f_1 & 0 \\ 0 & f_2 \end{pmatrix} \qquad K^y_{\alpha\beta} = \tfrac{1}{2}\partial_y T_{\alpha\beta} = 0. \quad K^z_{\alpha\beta} = \tfrac{1}{2}\partial_z T_{\alpha\beta} = 0.$$

Since both K tensors are zero equation (114) is a usual Killing equation $\nabla_{(\alpha}Z_{\beta)} = 0$ in 2 dimensions.

Case B.1 $f_1' \neq 0 \quad f_2' \neq 0$

The non-vanishing Christoffel symbols are

$$\Gamma^x_{xx} = \frac{f_2'}{2f_2} \qquad \Gamma^x_{tt} = -\frac{f_1'}{2f_2} \qquad \Gamma^t_{xt} = \frac{f_1'}{2f_1} \tag{128}$$

and the Ricci scalar is

$$R = \frac{f_1 f_1' f_2' + f_2(f_1'^2 - 2f_1 f_1'')}{2f_1^2 f_2^2} \tag{129}$$

Since we are in 2 dimensions, the Riemann tensor just has 1 independent component which is equal to the Ricci Scalar.

We use the property discussed in section (2) that for the case $rankT = 2$ we can obtain constrains on the 2 dimensional Killing vector field by calculating the Lie derivative of just the Ricci scalar. Knowing that for $f_1' \neq 0 \Leftrightarrow \partial_x R \neq 0$ we obtain

$$\pounds_Z R = Z^\mu \partial_\mu R = Z^x \partial_x R = 0 \qquad \Leftrightarrow Z^x = 0 \qquad Z_x = 0 \tag{130}$$

$$\begin{aligned} \pounds_Z T_{\mu\nu} = 0 \qquad &\Leftrightarrow \qquad \nabla_t Z_x + \nabla_x Z_t = 0 \\ \Leftrightarrow \partial_x Z_t - 2\Gamma^t_{xt} Z_t = 0 \qquad &\Leftrightarrow \qquad \partial_x Z_t = \frac{f_1'}{f_1} Z_t \end{aligned} \tag{131}$$

With the condition $\partial_t Z_t = 0$ and $c = constant$ we gain the Killing vector field

$$Z_\mu = \begin{pmatrix} f_1 c \\ 0 \end{pmatrix} \qquad Z^\mu = \begin{pmatrix} c \\ 0 \end{pmatrix} \tag{132}$$

and the collineation field $X = c\partial_t + \phi^y \partial_y + \phi^z \partial_z$ or

$$X^a = \begin{pmatrix} c \\ 0 \\ \phi^y \\ \phi^z \end{pmatrix} \qquad \text{and the collineation vector} \qquad X^1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \tag{133}$$

and an arbitray number of other collineation fields depending on the arbitrary functionsϕ^y, ϕ^z. Hence the Lie algebra is infinite.

Case B.2 $f_1' \neq 0 \qquad f_2' = 0$

This case is very similar to the previous one. Although the Ricci tensor has a different value, it is still non vanishing and eq. (130), (131) work as before. We arrive at the same collineation field as for the case $f_1' \neq 0 \qquad f_2' \neq 0$.

Case B.3 $f_1' = 0 \qquad f_2' = 0$

We get a usual Killing equation in 2 dimensions with 0 curvature.

$$Z^\mu = \begin{pmatrix} c_1 - cx \\ c_2 + ct \end{pmatrix} \qquad X^a = \begin{pmatrix} c_1 - cx \\ c_2 + ct \\ \phi^y \\ \phi^z \end{pmatrix}. \tag{134}$$

with the infinitesimal generators

$$X^1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \qquad X^2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \qquad X^3 = \begin{pmatrix} -x \\ t \\ 0 \\ 0 \end{pmatrix} \tag{135}$$

and an arbitray number of other collineation fields depending on the arbitrary functions ϕ^y, ϕ^z.

Case B.4 $f_1' = 0 \qquad f_2' \neq 0$

In this case the Riemann tensor (=Ricci tensor=Ricci scalar) vanishes and we arrive at the same collineation field as in the case $f_1' = 0 \qquad f_2' = 0$.

3.4 Rank 1

T_{ab} has $rank = 1$ under the condition that $f_1 = f_3 = 0$ or $f_2 = f_3 = 0$. Following the paper [6] the case Rank 1 can be treated very well with techniques coming from symplectic mechanics. Therefore we will give here a little theory about the Hamilton mechanics we will use.

3.4.1 Theory and a bit of Hamiltonian mechanics

Following the paper [6] we identify $rank = 1$ with a $T_{\mu\nu}$ that just consists of 1 component. It can be written as

$$T = \pm\theta \otimes \theta \tag{136}$$

with θ being a differential 1-form. The collineation condition eq. (3) is equivalent to the condition

$$\pounds_X \theta = 0 \tag{137}$$

Since θ is a differential form we can use **Cartan's magic formula** [12]

$$\pounds_X \theta = i_X d\theta + d(i_X\theta) = 0 \tag{138}$$

which relates the Lie derivative with the exterior and interior product.
This means that there exists locally a function ϕ such that

$$i_X d\theta = -d\phi \qquad\qquad i_X\theta = \phi \tag{139}$$

In Hamilton mechanics [13] one can find a special set of coordinates, called the canonical coordinates, that can be written as

$$\sum_i^2 p_i dq^i \tag{140}$$

They describe the complete state of a system with only 2 degrees of freedom, where the q_i are the configuration space coordinates and the p_i are their conjugate momenta, together they build the $2n$ dimensional phase space. The equations of motion of the Hamiltonian mechanics are

$$\frac{dq_k}{dt} = \frac{\partial H}{\partial p_k} \qquad\qquad \frac{dp_k}{dt} = -\frac{\partial H}{\partial q_k} \tag{141}$$

where H signifies the Hamiltonian. One can define the so called **symplectic form**

$$\omega = \sum_i dp_i \wedge dq^i = -\sum_i dq_i \wedge dp^i \tag{142}$$

which has the important property, that it defines a conserved quantity due to **Liouville's Theorem** [13].

It states that the phase space volume form on a symplectic manifold evolving due to the Hamiltonian equations of motion is preserved in time, i.e. it is invariant under the Hamiltonian flows. The Lie derivative of this volume form is zero along every Hamiltonian vector field.

Continuing with our 1-form θ and **Darboux Theorem** [14], we see that it implies the following theorem:

Let M be a 4-manifold and $\theta\epsilon\Lambda^1(M)$, there exist a canonical coordinate system $p_1, p_2, q^1.q^2$ a function Ψ, such that

$$\theta = d\Psi + e_1 p_1 dq^1 + e_2 p_2 dq^2 \qquad \{e_1, e_2 = 0, 1\} \qquad \{e_1 \geq e_2\}\,. \tag{143}$$

If we write X and $d\theta$ in canonical coordinates we have with the notation $\partial_i v = \frac{\partial v}{\partial q^i}$ and $\partial^i v = \frac{\partial v}{\partial p_i}$

$$X = X^i \partial_i + X_i \partial^i \qquad\qquad d\theta = \theta_i dq^i + \theta^i dp_i \tag{144}$$

With the first equation of (139) we formulate

$$-e_i X_i = \partial_i \phi \qquad\qquad e_i X^i = \partial^i \phi \tag{145}$$

From there, under the use of the second equation of (139) as well as equation (143) we get

$$X^i \theta_i + X_i \theta^i = \phi \qquad \theta_i := \partial_i \psi + e_i p_i \qquad \theta^i := \partial^i \psi \tag{146}$$

where different cases are possible depending on the value of e_1, e_2, see [6].

3.4.2 Case A: $f_1(x) \neq 0 \quad f_2(x) = 0$ and $f_3 = 0$

The rank=1 tensor T can be written as

$$T = f_1(x)dt^2 \qquad \text{with} \qquad \theta = \sqrt{|f_1(x)|}dt \tag{147}$$

being the differential 1-form.

• Case B.3 $f_1' \neq 0$

Following the paper [6] we are in the **[1.d.h.]** case, where

$$d\theta = \frac{f_1'}{s\sqrt{|f_1|}} dx \wedge dt \neq 0 \qquad \text{but} \qquad (d\theta)^2 = 0 \qquad \text{and} \qquad d\theta \wedge \theta = 0. \tag{148}$$

This means, following straightforward paper [6], that we can find a local chart such that $\theta = p_1 dq^1$, which we already wrote above $\theta = \sqrt{|f_1|}dt$, and

$$\phi - p_1 \partial^1 \phi = 0 \qquad \text{or} \qquad \phi = p_1 \Phi(q^1) \tag{149}$$

with Φ being an arbitrary function. This means in our particular case

$$p_1 = \sqrt{|f_1|}; \;\; q^1 = t; \;\; p_2 = y; \;\; q^2 = z$$

$$\phi - \sqrt{|f_1|}\partial^t \phi = 0 \qquad \text{or} \qquad \phi = \sqrt{|f_1|}\Phi(t) \tag{150}$$

The general collineation field reads

$$X = \Phi(q^1)\partial_1 - p_1 \Phi'(q^1)\partial^1 + X^2 \partial_2 + X_2 \partial^2 \tag{151}$$

which translates in our particular case to

$$X = \Phi(t)\partial_t - \frac{2f_1}{f_1'}\Phi'(t) + X^2 \partial_2 + X_2 \partial^2 \tag{152}$$

with the arbitrary functions $\Phi(t), X^2(t,x,y,z), X_2(t,x,y,z)$ and the fact that $p^1 = \sqrt{|f_1|}$ has been included.

• Case A.2 $f_1' = 0$

In this case we will rename $f_1 = c$ since it is just a constant. We have

$$d\theta = 0 \quad . \tag{153}$$

which means we are in the case **[1.d.0.]**. This means that there is a local chart $\{\tilde{t}, x, y, z\}$ such that $\theta = d\tilde{t}$. To realize this, we rename $d\tilde{t} = \sqrt{c}dt$ and find the collineation field

$$X = C\frac{\partial}{\partial t} + \sum_{\nu=2}^{4} X^{\nu}\frac{\partial}{\partial u^{\nu}} \tag{154}$$

with $X^{\nu}(\tilde{t}, x, y, z)$ being arbitrary and C=constant. The Lie algebra is infinite.

3.4.3 Case B: $f_2(x) \neq 0 \quad f_1(x) = 0$ and $f_3 = 0$

Analogously to the previous case, the rank=1 tensor T can be written as

$$T = f_2 dx^2 \qquad \text{with} \qquad \theta = \sqrt{f_2}dx \tag{155}$$

being the differential 1-form.

Again, we are in the case **[1.d.0.]** and following straight forward the paper [6], we get

$$d\theta = 0 \quad . \tag{156}$$

There exist a local chart $\{t, \tilde{x}, y, z\}$, such that $\theta = d\tilde{x}^1$. To realize this, we rename $d\tilde{x} = \sqrt{f_2}dx$ and find the collineation field

$$X = C\frac{\partial}{\partial \tilde{x}} + \sum_{\nu=2}^{4} X^{\nu}\frac{\partial}{\partial u^{\nu}} \tag{157}$$

with $X^{\nu}(t, x, y, z)$ being arbitrary and C=constant. Therefore the Lie algebra is infinite.

4 Conclusion

We found the Ricci collineations of a 2-covariant symmetric tensor following the procedure given in the paper [6]. In particular we did this in the example of the general static plane symmetric space time (14) where we analyzed the Ricci collineations depending on the rank of the Ricci tensor. We found that it gives a good technique to solve this problem.

For the non-degenerate case we obtain that there are 4 Ricci collineations. However, these are also isometries of the static plane symmetric metric and hence there are no proper Ricci collineations.

In the degenerate case we find that for $CaseB : f_2 = 0, f_1 \neq 0, \frac{f_3'}{f_3} = \tilde{f}$ in section (3.2.2) the Lie algebra is finite and there is a proper Ricci collineation that is not an isometry.

In all the other cases the Lie algebra of the collineation fields is infinite.

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